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The Editors
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Dear Editor,

I am pleased to submit my manuscript, “**Neighbourhood Structures and Ranking Operators for the Interval Job Shop Scheduling Problem**,” for consideration as a regular research article in *Applied Soft Computing*. The work sits squarely within the journal’s scope. It addresses scheduling under uncertainty modelled with intervals, a research line on fuzzy and interval job shop scheduling developed largely in this journal and its sister venues. Its methodology is a hybrid metaheuristic that embeds a tabu search within an artificial bee colony framework, with per-neighbourhood hyperparameter tuning by irace, and it is evaluated against published soft-computing solvers for the same problem.

The paper addresses the Interval JSP, where processing times are bounded but uncertain, and contributes on both theoretical and computational fronts. On the theoretical side, I formalise the notion of *extreme critical paths*, defined as the paths that are critical in the best-case or worst-case scenario graph, and use them to define a neighbourhood $H(\sigma)$ for local search. I prove that $H(\sigma)$ satisfies feasibility, connectivity and the no-loss-of-improving-neighbours property under any admissible interval ranking operator, and formulate five neighbourhood variants (N_1 , N_2 , N_3 , N_{ext} and a new variant N_8) within this framework. The interval semantics are made operational: a lemma shows that the interval makespan computed by propagation is exactly the range of realizable makespans, whose endpoints are the best- and worst-case makespans of the schedule. The LEX2 ranking therefore corresponds to lexicographic worst-case (robust) optimisation, which bridges the interval-ranking approach used in soft computing with the robust scheduling perspective of operations research. On the empirical side, the paper reports a 12,300-run study on 82 benchmark instances of up to 1000 operations, organised in three phases: a 5×4 comparison of four interval ranking operators (EV, LEX1, LEX2, YX), a head-to-head comparison of the five neighbourhoods after per-neighbourhood irace tuning of the hybrid solver, and a comparison against three published IJSP solvers (GA, *fEABC*, ESABC). The tuned hybrid with neighbourhood N_2 reduces mean relative error by $\approx 57\%$ over the previous best published solver on every one of the 82 instances. A float analysis further explains, in structural terms, why LEX2 yields consistently narrower makespan intervals and thus tighter uncertainty guarantees.

In line with the journal’s emphasis on reproducibility, all raw run logs, aggregated CSVs, irace tuning records, instance files and analysis scripts are openly available on Zenodo ([10.5281/zenodo.20562888](https://zenodo.org/record/20562888)), with per-instance results on the 70 Taillard instances in a separate supplementary document ([10.5281/zenodo.20562857](https://zenodo.org/record/20562857)). I confirm that this manuscript has not been published previously, is not under consideration by any other journal, and that all listed contributors have approved the submission. I declare no competing financial interests or personal relationships that could have appeared to influence the work reported. Thank you for considering this submission.

Sincerely,

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